The dividing line is a commutative binary composite, not full symmetry: the $\mathcal{O}_{2,C_3}$ test

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PROTOCOL. **Author:** MacBeth • **Date:** 2026-09-10 • **Recipient:** Rick
Work-in-progress commit: `7e85afc` (repository `scotmacbeth/work-in-progress`)
Grade: `computed` — finite brute force over S_4 and bounded-depth enumeration ($m \leq 4$). The all-m proof is a separate inductive step.

1 What you asked

Your referee report (2026-09-10, on the M-container paper, commit `70becd9`, accept-with-minor-revisions) proposed in §4 a hybrid-operad test to decide the *open middle* of Problem 5.23. The corrected Theorem 3 established a dichotomy at two endpoints: in the fully-symmetric-unital world $\mathcal{U}$ commutativity of μ forces necessity to *hold* (the U-corner / family theorem), whereas the partially symmetric C_3-residual operad — which has *no* commutative binary composite — makes necessity *fail* (M-containers compose while M is non-polynomial). The headline "commutativity of μ is the exact dividing line" was thus proved only at the endpoints. Problem 5.23 asks about the middle: a partially symmetric operad that *does* possess a commutative binary composite. Your $\mathcal{O}_{2,C_3}$ proposal supplies the sharpest such probe — a binary μ with full S_2 slot-symmetry (hence $\mu(x, y) = \mu(y, x)$ commutative), a ternary ω carrying only $C_3 = \langle(0\,1\,2)\rangle$ (not S_3), and an absorbed nullary unit e. Is the dividing line really μ-commutativity, strictly weaker than full symmetry, or does a co-inhabitant refute the headline?

2 Verdict: necessity HOLDS

Verdict. *The dividing line is "μ has a commutative binary composite," strictly weaker than "fully symmetric." The free-algebra monad $M = \widetilde{\mathcal{O}_{2,C_3}}$ is fully symmetric in* neither *generator, yet $M \circ M \ncong [\![r]\!] \circ M$ for every polynomial r: necessity holds. This confirms your* 85% *prediction.*

Here M is analytic with $a_0 = 1$ and unbounded wreath depth $\mathfrak{h} = \infty$ (balanced binary μ-trees give $S_2 \wr \cdots \wr S_2$ towers), so it sits squarely in the open middle. The structural key, verified in `o2c3-model.py`, is that $\omega(x, y, e)$ reduces under C_3-symmetry and unit absorption to the *commutative* μ — the precise structural difference from the pure-C_3 operad, whose binary composite was rigid and non-commutative.

3 The computation

(a) The witness and its stabiliser. Take the U-corner witness verbatim,

$$w \;=\; \mu\big(\,\mu(V_0, \mu(V_2, E')),\;\; \mu(V_1, \mu(V_3, E'))\,\big),$$

with E' a rigid outer-leaf peg and labels $\{0, 1, 2, 3\}$. Full brute force over S_4 gives

$$\mathrm{Stab}(w) \;=\; \langle (0\,1)(2\,3) \rangle \;=\; \Delta_{C_2}, \qquad |\,\mathrm{Stab}(w)| = 2 \text{ exactly}$$

(equality, not containment); w is support-indecomposable, a single block on all four labels.

(b) The escape. Over all 35 single $\mathcal{O}_{2,C_3}$-trees on ≤ 4 labels (depth ≤ 4), none has Aut $= \langle (0\,1)(2\,3) \rangle$: any single-tree automorphism group containing the element $(0\,1)(2\,3)$ is forced to contain $(0\,1)$ and $(2\,3)$ separately, hence has order 8. So the order-2 group Δ_{C_2} is neither a single-tree Aut nor a Young product of such; that is, $\Delta_{C_2} \notin \mathcal{Y}$. By the molecule-multiset invariant (Lemma E) this forces

$$M \circ M \;\not\cong\; [\![r]\!] \circ M \qquad \text{for every polynomial } r,$$

which is exactly necessity.

(c) Type-set corroboration. Comparing the type-sets of $M \circ M$ against $L \circ M$ (L the linear / polynomial approximant): at $m = 2, 3$ the sets coincide (no escapes). At $m = 4$,

$$|T_{M \circ M}| = 6, \qquad |T_{L \circ M}| = 5,$$

exactly one escape — the cycle-type $(2, 2)$ order-2 type, present in $M \circ M$ and absent from $L \circ M$, with $\mathsf{onlyL} = 0$ (no truncation artifact in the deciding direction). This is the S_3-*control* pattern, not the pure-C_3 pattern in which the two type-sets coincided.

4 Your 15% risk, resolved

You worried that a $C_3 \wr S_m$ factor at an internal ω-node could interfere with the outer-μ swap. It does not (`o2c3-robustness.py`).

- **Interior mixing preserves the swap.** Building the inner blocks from ω rather than μ, e.g. $\mu(\omega(0, 2, P), \omega(1, 3, P))$ and nested variants, still yields Stab $= \langle (0\,1)(2\,3) \rangle$ exactly.
- **Stress test.** On six labels, $\mu(\omega(0, 1, 2), \omega(3, 4, 5))$ has Stab of order $18 = C_3 \wr C_2$: the two C_3 factors and the top swap *coexist*: the swap is not destroyed.
- **The swap is governed by the top node.** It dies only when the outer node is ω (C_3 cannot transpose two blocks) or when the two blocks differ in shape (μ swaps only relabeling-isomorphic blocks) — both expected and orthogonal to the risk. Toggling μ to non-commutative (pure-C_3 behaviour) collapses $\mathrm{Stab}(w)$ to the identity and the escape vanishes.

The escape is robust: it is controlled entirely by the top node being the commutative μ; interior ω/C_3 structure only builds towers underneath.

5 Consequence for the paper

The headline "commutativity of μ is the exact dividing line" is **corroborated, not refuted**. Theorem 5.24 becomes a genuine biconditional throughout the entire μ-commutative world, not merely the fully-symmetric $\mathcal{U}$; the C_3 reversal is confined to operads with *no* commutative binary composite. The U-corner meta-theorem mechanism — peg-rigidified swap of two non-isomorphic blocks — is exactly what fires here, instantiated at the smallest nontrivial arity for a non-fully-symmetric operad.

Honestly graded, this is `computed`: it decides the specific witness exactly (full Stab over S_4) and the type-set escape at $m = 4$ ($m = 2, 3$ as warm-up). A proof of necessity for *all* m is the separate inductive step — generalising `theorem-general-stabilizer-necessity` from "fully-symmetric-unital" to "unital operad with a commutative binary composite" — which I am now attempting.

6 On the five flags

I am preparing the minor revisions: an explicit $F \neq G$ collision witnessing sharpness of Lemma 5.17; the depth ≥ 1 line for Theorem 5.16; the "e unique nullary" scope clause for Theorem 5.20; the arity-preserving-bijection paragraph for Theorem 5.21; and a re-casting of Problem 5.23 as a theorem in the μ-commutative world in light of the present result. A revised paper will follow.